

STACK

Q7) Next Greater Element and Next Smaller Element

DS	Session	Stack	Question Information	Level 1	Challenge 47
Question description You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined: $F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$ $G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$ Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i, output -1 as its answer. If such a number does exist, output $A[G(F(i))]$ Constraints: $1 \leq N \leq 3000$ $0 \leq A[i] \leq 10^{18}$ Input : The first line contains a single integer N denoting the size of array A. Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$. Output : Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or -1, if $G(F(i))$ does not exist.					

CODE:

```

C main.c > main
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main(void) {
5      int n;
6      if (scanf(format: "%d", &n) != 1) return 0;
7
8      long long *a = (long long *)malloc(size: n * sizeof(long long));
9      for (int i = 0; i < n; i++) {
10         scanf(format: "%lld", &a[i]);
11     }
12
13     int *next_greater = (int *)malloc(size: n * sizeof(int));
14     int *next_smaller = (int *)malloc(size: n * sizeof(int));
15     int *stack = (int *)malloc(size: n * sizeof(int));
16     int top = -1;
17
18     for (int i = n - 1; i >= 0; i--) {
19         while (top >= 0 && a[stack[top]] <= a[i]) {
20             top--;
21         }
22         next_greater[i] = (top == -1) ? -1 : stack[top];
23         stack[++top] = i;
24     }
25
26     top = -1;
27     for (int i = n - 1; i >= 0; i--) {
28         while (top >= 0 && a[stack[top]] >= a[i]) {
29             top--;
30         }
31         next_smaller[i] = (top == -1) ? -1 : stack[top];
32         stack[++top] = i;
33     }
34
35     for (int i = 0; i < n; i++) {
36         int fg = next_greater[i];
37         if (fg != -1 && next_smaller[fg] != -1) {
38             printf(format: "%lld ", a[next_smaller[fg]]);
39         } else {
40             printf(format: "-1 ");
41         }
42     }
43     printf(format: "\n");
44
45     free(ptr: a);
46     free(ptr: next_greater);
47     free(ptr: next_smaller);
48     free(ptr: stack);
49     return 0;
50 }

```

OUTPUT:

```
[suraj@suraj-83gs test]$ ./main
```

```
8
```

```
3
```

```
7
```

```
1
```

```
7
```

```
8
```

```
4
```

```
5
```

```
2
```

```
1 4 4 4 -1 2 -1 -1
```

```
[suraj@suraj-83gs test]$
```